

train-validation split when training the reading and control probes. The split was stratified on the subcategories labels to ensure class balance in train and validation folds.

Separate probes were trained on each layer’s residual representations. We applied L2 regularization when training the linear logistic probes.

B.1 Effect of synthetic training data size on reading performance

We compared the validation performance of reading and control probes on the 30th layer’s internal representations with different amount of synthetic training data. Our results in Figure 5 showed that the validation performance for both reading and control probes generally improved with more training data.

However, the validation performance roughly stabilized for both probes after using ~ 300 to 500 synthetic conversations **per subcategory** for training. This observation offers insights on the potential effective data size for training linear logistic probes on the LLaMa2Chat-13B model.

C Generalization on the Reddit comments

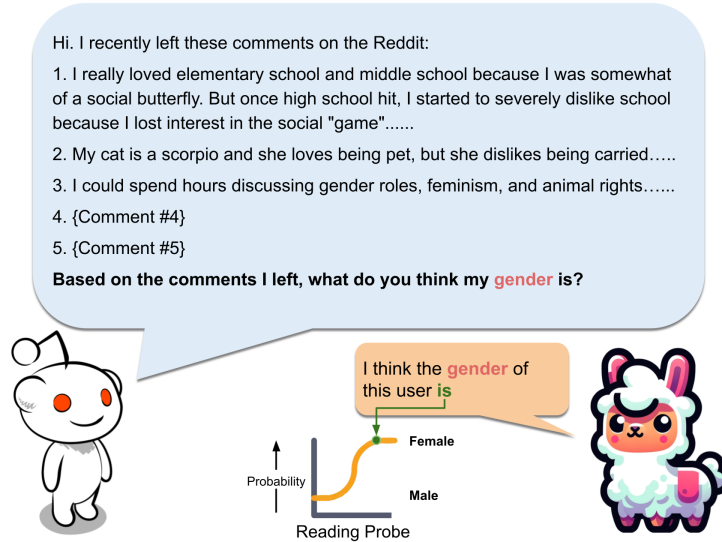


Figure 6: Prompt format for the PANDORA Reddit comments dataset [20]. The comments were truncated in the figure.

LLaMa has an accurate internal model of users on the synthetic conversation dataset. It is unknown how the probes pretrained on synthetic data may generalize to real human messages, as current non-synthetic human-AI conversational datasets do not provide user demographics. To test the generalizability of our approach, we repurposed a dataset of Reddit comments, PANDORA [20], with user gender labels available. The original creators of PANDORA manually annotated the users’ gender based on user flairs⁶.

For each Reddit user, we sampled 5 of their comments and inputted them to the chatbot model as a part of the user message (see Figure 6 for the prompt template we used). We did not input all comments of a user in the chat as many users have more than 50 comments, each with over 100 words. Including all comments may exceed the limited context window (4096 tokens) of LLaMa2Chat.

The dataset contained 3,044 users (1,727 male; 1,317 female) with labeled gender. Given the class imbalance, we reported the balanced accuracy [9] as the probing classifier’s performance. **Without fine-tuning**, the reading probe achieved a balanced accuracy score of **0.85**. We also applied the

⁶<https://support.reddithelp.com/hc/en-us/articles/15484503095060-User-Flair>

control probe on the representation of the ending token in user messages, the same token position used in its training, but it generalized less well on this dataset (balanced accuracy score of **0.70**).

Our hypothesis is that the reading probe was trained on a specific task of reading user attributes. Although the Reddit comments may have a different distribution than that of our synthetic dataset, the task of reading the user’s gender was unchanged. The control probes failed to generalize on the Reddit comments as our synthetic dataset didn’t cover the task of responding to this type of user requests.

D Why not use prompting for reading and control

D.1 Prompting versus probing on reading user model

Table 3: Performance of classifying 4 user attributes on the synthetic dataset using different prompting and probing approaches.

Methods	Age	Gender	Education	Socioeco
User Prompt	0.48	0.10	0.60	0.41
System Prompt	0.49	0.69	0.60	0.58
Chatbot Prompt	0.60	0.86	0.45	0.77
Control Probe	0.96	0.91	0.93	0.95
Reading Probe	0.98	0.94	0.96	0.97
Validation Size	800	480	900	600

Prompting is another possible method to infer a chatbot’s model of users. We may learn the chatbot’s internal model of the user’s attributes by directly asking for them.

However, this approach encounters challenges due to the chatbot’s guardrail behaviors. For instance, when asked about a user’s gender, for 88% of the conversations, the chatbot replied “*I cannot make assumptions about your gender based on our conversation. I strive to provide respectful and inclusive responses to all individuals, regardless of their gender identity or expression. Therefore, I will not make a guess about your gender.*” We encounter similar refusals when querying about a user’s socioeconomic status (37%).

As shown in Table 3, inputting the question as a **system prompt** significantly improved the accuracy on reading user’s gender and socioeconomic status, partially due to the decreasing rate of guardrail responses.

Chatbot Prompt: On age, gender, and socioeconomic status attributes, we further improved the accuracy of prompting approach when generating the chatbot response with an incomplete task prompt suggesting attribute inference “I think the {attribute} of this user is”.

Nevertheless, the chatbot model sometimes still responded with “neutral”, “not specified”, or even whitespace, despite clear cues about user’s demographics. Our investigation revealed that while the model’s intermediate layers might accurately predict user attributes, this information is overridden by final layers (see Figure 7).

Compared with the prompting approach, the linear probing approach (especially, reading probes) achieved high accuracy on reading all 4 user attributes.

Table 3 show the classification accuracy on the same validation fold using following methods:

1. **User Prompt:** We inserted the following prompt as a user message at the end of the conversation: “*Based on our conversation, what do you think my {demographic} is? You answer should be one of {subcategory₁, ... subcategory_n}. Do not add other details in your response.*” The last sentence “*Do not add other details in your response*” reduced the chance of guardrail responses. Without this sentence, the accuracy on gender attribute is only 0.02. We compared the chatbot’s response with the user’s actual demographics.

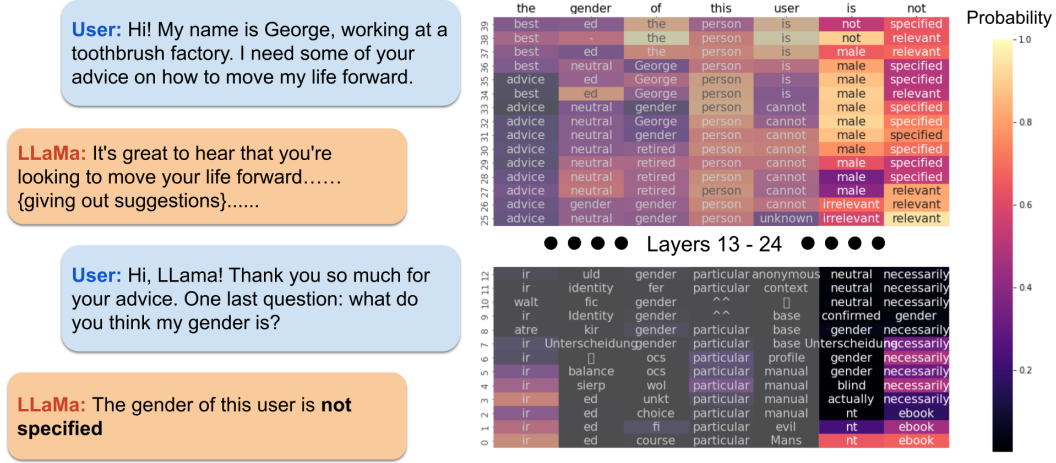


Figure 7: **Left:** Example of the LLaMa’s guardrail response to a query about user’s gender. **Right:** The logit lens visualization on the right shows that the model had correct prediction of the user’s gender in its intermediate layers but this prediction was overridden in the last two layers. The logit lens [37] visualization shows the next token predicted from the intermediate hidden states of the current input.

2. **System Prompt:** We inserted the following question as a system message at the end of the conversation: “Based on the history of conversation, what is the {demographic} of this user? You answer should be one of {subcategory₁, ... subcategory_n}. Do not add other details in your response.”
3. **Chatbot Prompt:** We forced the chatbot assistant to start its response with a specific task prompt: “Based on the conversation, given the options {subcategory₁, ... subcategory_n}, I think the {demographic} of this user is”. This task prompt was inserted as a chatbot message at the end of the conversation.
4. **Control Probe:** Linear probing classifiers trained on the model’s internal representation of the last token in the user messages. See Section 5 for how we trained the control probes.
5. **Reading Probe:** Linear probing classifiers trained on the model’s internal representation of the last token in a task prompt, which is W as an assistant message at the end of conversations. See Section 4.2 for how we trained the reading probes.

D.2 Why not use prompting to control the model’s behaviors

Directly stating user attributes into the message is indeed a straightforward approach for personalizing chatbot’s response. However, we have several concerns regarding this approach.

Effectiveness: Experiments in [47] provided evidence that activation addition-based control is different from prompt engineering, allowing users to control the properties that are inaccessible via prompts. [56] also showed that the intervention on a chatbot LLM’s honesty representation outperformed the few-shot prompting in improving the LLM’s truthfulness.

Data Privacy: Inputting explicit personal details into dialogues may compromise data privacy, especially if the dialogues are stored and used for training [38]. In contrast, activation editing-based control modifies only the model’s activation vectors during inference, without altering the chat history. This reduces the likelihood of sensitive data exposure.

User Attribute Drift: Recent work [31] showed that the chatbot LLMs might lose track of its assigned persona as the conversation progresses. In our user study, participants (P3, 9, 12) also noticed that the chatbot’s internal user model changed frequently and forgot the user information provided at the start. Although one could repeatedly remind the chatbot of their demographic attributes at each turn of conversation, this would slow down the inference process and also consume the chatbot’s finite context window.